

## EXPERIMENT NO : 07

**AIM: To study & Implement Security as a Service using IAM service.**

### Objective

1. Understand the concept of Security as a Service (SECaaS)
2. Learn IAM components (Users, Groups, Roles, Policies)
3. Implement access control using IAM
4. Apply the Principle of Least Privilege
5. Test and verify permissions

### \*INTRODUCTION:

#### Security as a Service (SECaaS)

Security as a Service is a cloud-based security model where security services such as authentication, authorization, identity management, and monitoring are provided by a cloud provider.

**Security as a Service (SECaaS)** is a cloud-based model where security services such as identity management, access control, threat detection, data protection, and compliance monitoring are delivered by a cloud provider instead of being managed on-premises.

In **Amazon Web Services (AWS)**, security is delivered as a shared responsibility between AWS and the customer. AWS secures the cloud infrastructure, while customers secure their data, users, and configurations.

AWS provides multiple managed security services that function as SECaaS solutions, reducing operational complexity and improving protection.

#### Advantages of Security as a Service (SECaaS)

1. **Cost Efficiency** – Reduces capital expenses by eliminating the need for on-premises security hardware and maintenance.
2. **Scalability** – Easily scales security services up or down based on business requirements.
3. **Centralized Management** – Provides a unified platform to monitor and manage security policies and controls.
4. **Automatic Updates** – Ensures continuous protection through automatic security patches and updates.
5. **High Availability** – Offers reliable and resilient security infrastructure with minimal downtime.
6. **Improved Compliance** – Helps organizations meet regulatory and industry security standards.

## Applications of AWS Security as a Service (SECaaS)

1. **Identity and Access Management** – Controls user authentication and authorization to securely access AWS resources.
2. **Threat Detection and Monitoring** – Continuously monitors systems to detect and respond to security threats in real time.
3. **Data Protection and Encryption** – Safeguards sensitive data using encryption and secure key management solutions.
4. **Network Security** – Protects applications and infrastructure from cyberattacks such as DDoS and unauthorized access.
5. **Compliance Management** – Helps organizations meet regulatory requirements through auditing and reporting tools.
6. **Application Security** – Secures web applications and APIs from common vulnerabilities and malicious traffic.

### \* Identity and Access Management (IAM)

**AWS Identity and Access Management (IAM)** is a security service provided by **Amazon Web Services (AWS)** that enables you to securely control access to AWS resources.

IAM allows administrators to create and manage **users, groups, roles, and policies** to define who can access specific AWS services and what actions they can perform. It is a core component of AWS security and follows the **principle of least privilege**, meaning users get only the permissions they need.

IAM is provided at no additional cost and is used to manage authentication (who you are) and authorization (what you can do).

### Advantages of IAM

1. **Secure Access Control** – Provides fine-grained control over AWS resources.
2. **Free to Use** – No extra charges for IAM service.
3. **Granular Permissions** – Allows detailed permission settings for specific services and actions.
4. **Supports MFA** – Enhances account security with multi-factor authentication.
5. **Temporary Credentials** – Uses roles for secure temporary access instead of permanent keys.
6. **Centralized Management** – Manage all users and permissions from one place.
7. **Integration with AWS Services** – Works seamlessly with all AWS services.

### Applications of IAM

1. **Enterprise Access Management** – Manage employee access to AWS resources securely.
2. **Application Authentication** – Allow applications to securely access AWS services using roles.
3. **Secure API Access** – Control programmatic access using access keys.

4. **Cross-Account Access** – Grant access between different AWS accounts securely.
5. **Role-Based Access Control (RBAC)** – Assign permissions based on job roles.
6. **Cloud Resource Protection** – Restrict unauthorized access to critical resources.

## Uses of IAM in Real-World Scenarios

- IT administrators controlling access to cloud infrastructure
- Developers accessing specific services like S3 or EC2
- EC2 instances accessing other AWS services securely using roles
- Organizations implementing security policies and compliance requirements

## Key Components of IAM

1. **Users:** In AWS Identity and Access Management (IAM), a User represents an individual person or an application that needs access to AWS resources

**Example:** An employee named Rahul is given an IAM user account to access only Amazon S3 for uploading files.

2. **Groups:** A Group is a collection of IAM users who share the same permissions.

**Example:** A group named Developers may have permission to access EC2 and S3 services, and all developer users are added to this group.

3. **Roles:** A Role is an IAM identity that provides temporary access permissions to AWS services, users, or external entities.

**Example:** An EC2 instance is assigned a role that allows it to read data from S3 without storing access keys.

4. **Policies:** A Policy is a JSON document that defines permissions.

**Example Policy:** Allows read-only access to S3 buckets. Policies are the core mechanism that defines authorization in IAM.

5. **MFA (Multi-Factor Authentication):** Multi-Factor Authentication (MFA) adds an extra layer of security to user login.

**Example:** Even if someone steals a password, they cannot log in without the OTP code.

## Conclusion :

This experiment demonstrated how AWS Identity and Access Management (IAM) in Amazon Web Services provides Security as a Service by managing authentication and authorization. By creating users, roles, and policies, we implemented secure and controlled access to AWS resources following the principle of least privilege.